NEW LIGHT
INSTITUTE
Medical | Foundation

• Test ID : 2

ALL INDIA
FULL SYL

DURATION : 180 Minutes **DATE : 0**

Download our App
New Light | NEET Prep

Topic (

Physics	**: FULL SYLLABU**
Chemistry	**: FULL SYLLABU**
Biology	**: FULL SYLLABU**

(Do not open this Test Bookle

Please read the instructions carefully :

1. The test is of **3 hours** duration and Test Booklet contains response, the candidate will get **4 marks**. For each incor are **720**.
2. Use **Blue/Black Ball Point Pen** only for writing particula
3. Rough work is to be done on the space provided in the Te
4. **On completion of the test, the candidate must handover the candidates are allowed to take away this Test Booklet wit**
5. The candidates should ensure that the Answer Sheet is not write your roll no. anywhere else except in the specified s
6. Before attempting the question paper ensure that it contai
7. Each candidate must show on demand his/her Admission
8. If any student is found to have occupied the seat of another and shall have to accept any other penalty imposed upon
9. No candidate, without special permission of the Superinte
10. The candidates should not leave the Examination Hall with sign the Attendance Sheet twice.
11. Use of Electronic/Manual Calculator is prohibited.
12. The candidates are governed by all Rules and Regulations All cases of unfair means will be dealt with as per Rules ar
13. The candidates will write the Correct Test ID Code as giv

Name of the Student (In CAPITALS) : _____

Candidate ID : _____

Candidate Signature : _____

INSTRU

- The candidates should ensure that the Answer Sheet is not f write your roll no. anywhere else except in the specified sp
- Before attempting the question paper ensure that it contain
- Each candidate must show on demand his/her Admission C
- If any student is found to have occupied the seat of another and shall have to accept any other penalty imposed upon t
- No candidate, without special permission of the Superinten
- The candidates should not leave the Examination Hall witho sign the Attendance Sheet twice. Cases where a candidate not to have handed over Answer Sheet and dealt with as a
- Use of Electronic/Manual Calculator is prohibited.
- The candidates are governed by all Rules and Regulations o All cases of unfair means will be dealt with as per Rules an
- The candidates will write the Correct Test ID Code as give

Key Points of New Light Test

- Rapid Fire Revision of all tests Live C
  Channel (YouTube) before the sch
- Video of all tests' solution available
- Chat support **24×7** available for the
- Test results are regularly sent to the
  For lattest update on NEET, PDF shee

Please Subscribe our –

Telegram Channel - @New

Youtube Channel - New L

For Today's Paper Discussion - Sc

• Youtube Channel Link :

https://www.youtube.com/@newlight

• Youtube Channel Name : New Ligh

BEWARE OF NEG

TOPIC : FULL SYLLABUS

1. A wheel has angular acceleration of 3.0 rad/sec^2 and an initial angular speed of 2 rad/sec. In a time of 2 sec it has rotated through an angle (in radian) of .....

   (1) 10

   (2) 12

   (3) 4

   (4) 6.

2. On the horizontal surface of a truck (μ = 0.6); a block of mass 1 kg is placed. If the truck is accelerating at the rate of 5m/sec^2 then acceleration (in m/s^2) of the block with respect to truck will be:

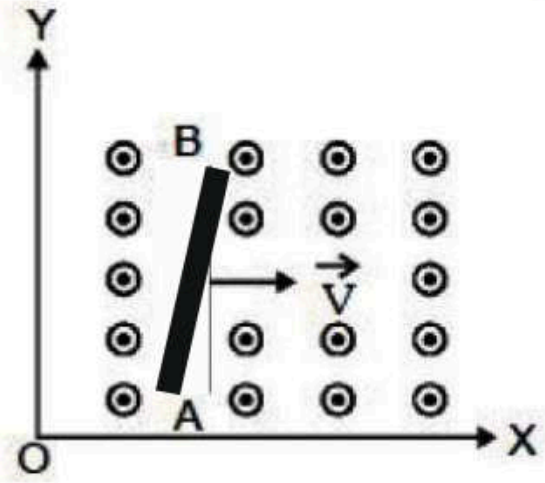
   (1) 5

   (2) 0

   (3) 1

   (4) 2

3. A conducting rod AB moves parallel to X-axis in a uniform magnetic field, pointing in the positive X. direction. The end A of the rod gets



   (1) positively charged

   (2) negatively charged

   (3) neutral

   (4) first positively charged and then negatively charged

NEW LIGHT INSTITUTE All India Full Syllabus Test–31 [Paper]

4. Red light of wavelength 5400 Å from a distant source falls on a slit 0.80 mm wide. Calculate the percentage increase in distance between first two dark bands on each side of central bright band in the diffraction pattern observed on a screen place 1.4m from the slit if wavelength is changed to 6000 Å:

   (1) 11.1

   (2) 0

   (3) -11.1

   (4) 5

5. If an electron in a hydrogen atom jumps from the 3rd orbit to the 2nd orbit, it emits a photon of wavelength λ. when it jumps from the 4th orbit to the 3rd orbit, the corresponding wavelength of the photon will be :

   (1) $\frac{20}{7}\lambda$

   (2) $\frac{20}{13}\lambda$

   (3) $\frac{16}{25}\lambda$

   (4) $\frac{9}{16}\lambda$

6. If $|\vec{A}\times\vec{B}| = \sqrt{3}|\vec{A}\cdot\vec{B}|$ then the angle between $\vec{A}$ and $\vec{B}$ can be :

   (1) $2\pi/3$

   (2) zero

   (3) $\pi/3$

   (4) Both 1 and 3

7. A wave of frequency 100 Hz is sent along a string towards a fixed end. When this wave travels back after reflection, a node is formed at a distance of 10 cm from the fixed end of the string. The speed of incident (and reflected) wave are in m/s :

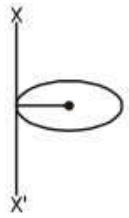
   (1) 40

   (2) 20

   (3) 10

   (4) 5

8. A wire of length L and uniform linear density m shaped a circular loop as shown in figure. centre of loop is O. The moment of inertia of loop about axis xx' is : xx' is perpendicular to loop and passing through Rim:



(1) $\frac{2mL^3}{4\pi^2}$ (2) $\frac{3mL^3}{16\pi^2}$

(3) $\frac{3mL^3}{8\pi^2}$ (4) $\frac{mL^3}{16\pi^2}$

9. For a given angle of the projectile if the initial velocity is doubled the ratio of the range and maximum height of the projectile becomes will be :

(1) Halved

(2) Quarterpuled

(3) Doubled

(4) Remain same

10. The displacement of a body of mass 1 kg varies with time t as $S = 3t^2 + 2t + 1$, where t is in seconds. The work done by all the forces acting on the body during the time interval t = 2 sec to t = 4 sec is :

(1) 36 J

(2) 64 J

(3) 240 J

(4) 480 J

11. Two moles of a monoatomic ideal gas is confined in a container and is cooled such that its temperature is decreased by 10°C. The approximate change in its internal energy is ..... J. (R = 8.31 J/mole .. K)

(1) +250

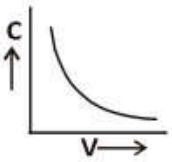
(2) +350

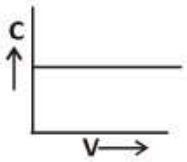
(3) –250

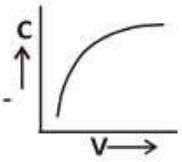
(4) +450

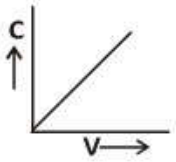
NEW LIGHT INSTITUTE All India Full Syllabus Test–31 [Paper]

12. Variation of potential V with capacitance C is best given by :

(1) 

(2) 

(3) 

(4) 

13. Imagine earth to be a solid sphece of mass M and radius R. If the value of acceleration due to gravity at a point A depth d below earth's surface is same as its value at point B height h above its surface and equal to g/4 (where g is the value of acceleration due to gravity on the surface of earth), the distance between A and B :

(1) 7R/4

(2) 3R/2

(3) 7R/3

(4) 2R

14. Four lead-acid type of secondary cells each of emf 3.0 V and internal resistance 0.15 ohm are joined in series to provide a supply to a resistance of 5.4 ohm. Find termial voltage of a perticular cell:

(1) 0 V

(2) 8.2 V

(3) 2.7 V

(4) 10.8 V

15. Magnetic moment of toroid is :

(1) Infinity

(2) Zero

(3) Finite, very small

(4) Depends on size.

16. The average kinetic energy of a gas molecule at 27°C is 6.21×10^{-21} J. Its average kinetic energy at 227°C will be :

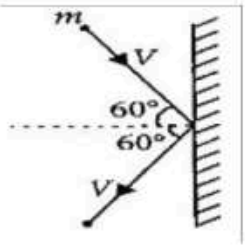
(1) 52.2×10^{-21} J

(2) 5.22×10^{-21} J

(3) 10.35×10^{-21} J

(4) 11.35×10^{-21} J

17. A rigid ball of mass m strikes a rigid wall at 60° and gets reflected without loss of speed as shown in the figure. The value of impulse imparted by the wall on the ball will be :



(1) $\frac{mv}{2}$

(2) $\frac{mv}{3}$

(3) mv

(4) 2mv

18. The time period of a satellite of earth is 5 hours. If the separation between the earth and the satellite is increased to four times the previous value, the new time period will become 10x. The value of x is:

(1) 2

(2) 1

(3) 8

(4) 4

19. A simple harmonic oscillator has an amplitude a and time period T. The time required by it to travel from x = a to x = - $\frac{a}{2}$ is

(1) T/6

(2) T/4

(3) T/3

(4) T/2

20. Two similar rods but of different materials are joined together end to end. The ratio of their thermal conductivities is 3 : 4. The composite rod is lagged between melting ice at 0°C and steam at normal pressure. The temperature difference between mid point of the rod :

(1) 60°C

(2) 50°C

(3) 40°C

(4) 57.14°C.

NEW LIGHT INSTITUTE All India Full Syllabus Test–31 [Paper]

21. $V = k\left(\frac{P}{T}\right)^{0.33}$ where k is constant, it is :

(1) isothermal process

(2) adiabatic process

(3) isochoric process

(4) isobaric process

22. The position x of a particle with respect to time t along x..axis is given by $x = 9t^2 - t^3$ where x is in metres and t in seconds. What will be the position of this particle when it achieves maximum speed along the x direction?.....m

(1) 54

(2) 81

(3) 24

(4) 32

23. A photosensitive metallic surface has work function, hv_0. If photon of energy $4hv_0$ fall on this surface, the electrons come out with a maximum velocity of 9×10^6 m/s. When the photon energy is increased to $8hv_0$, then maximum velocity of photoelectrons will be :

(1) 13.7×10^6 m/s

(2) 9.7×10^6 m/s

(3) 11.7×10^7 m/s

(4) 16.7×10^5 m/s.

24. If ${}_3Li^7 + {}_1H^1 \rightarrow {}_4Be^8$ mass of ${}_3Li^7$ = 7.0020 u mass of ${}_1H^1$ is 1.0027 u and mass of ${}_4Be^8$ is 8.0016u. Then energy evolved is :

(1) 2.89 cal

(2) 2.89 eV

(3) 2.89 MeV

(4) 1.86 MeV.

25. A person is standing in an elevator. In which situation he finds his weight less than actual when

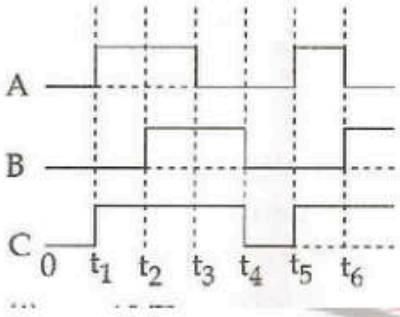
(1) The elevator moves upward with constant acceleration

(2) The elevator moves downward with constant acceleration.

(3) The elevator moves upward with uniform velocity

(4) The elevator moves downward with uniform velocity

26. The figures shows a logic circuit with two inputs A and B and the output C. The voltage wave forms across A, B and C are as given. The logic circuit gate is :



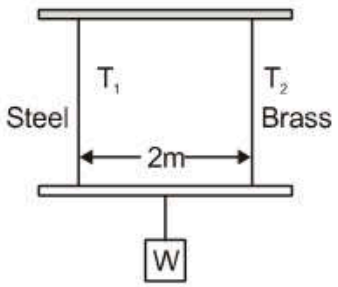
(1) AND gate

(2) NAND gate

(3) OR gate

(4) NOR gate

27. A light rod of length 2m suspended from the ceiling horizontally by means of two vertical wires of equal length. A weight W is hung from a light rod as shown in figure. The rod hung by means of a steel wire of cross-sectional area $A_1 = 0.4\ cm^2$ and brass wire of cross-sectional are $A_2 = 0.6\ cm^2$. To have equal stress in both wires, $T_1 / T_2 =$



(1) 1/3

(2) 1/4

(3) 4/3

(4) 2/3.

28. If the work done in blowing soap bubble of volume V is W, then the work done in blowing a soap bubble of volume 8V will be :

(1) W

(2) 4W

(3) 6 W

(4) $W(4)^{1/3}$

29. If temperature of a black body increases from 7°C to 287°C, then the rate of energy radiation increases by :

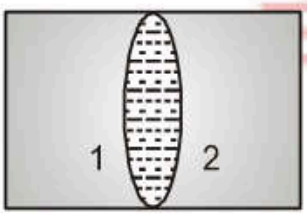
(1) $\left(\frac{287}{7}\right)^4$ times

(2) 16 times

(3) 15 times

(4) 2 times

30. Two plano-concave lenses (1 and 2) of glass of refractive index 1.5 have radii of curvature 20 cm and 15 cm. They are placed in contact with their curved surface towards each other and the space between them is filled with liquid of refractive index 3/2. Then the combination is :



(1) Convex lense

(2) Concave lense

(3) Sheet

(4) Prism

31. In Young's double slits experiment carried out with light of wavelength 6000A, the distance between the slits is 0.4mm and the screen is at 2 meter from the slits. Position of 3rd maximum from centre of screen is -

(1) 0.09 cm

(2) 5 cm

(3) 0.08 cm

(4) 0.02 cm

32. **In an interference experiment, 3rd bright fringe is obtained on the screen with a light of 700 nm. What should be the wavelength of the light source in order to obtain 5th bright fringe at the same point:**

(1) 420 nm (2) 500 nm

(3) 750 nm (4) 630 nm.

33. **From the top of tower 3 bodies are thrown simultaneously**

I. 15 kg vertically up with 20 m/s

II. 10 kg with 20 m/s 60^0 above north horizon

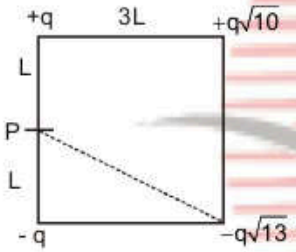
III. 20 kg with 10 m/s 30^0 above east horizon

Acceleration of the centre of mass of three particle system will be

(1) $g/2$ (2) g

(3) $2g$ (4) $g/3$

34. **Find potential at point P :**



(1) $\frac{q}{8\pi\varepsilon_0 L}$ (2) $\frac{q}{4\pi\varepsilon_0 L}$

(3) $-\frac{q}{4\pi\varepsilon_0 L}$ (4) zero

35. **Electric field associated with EM wave in free space is given by $\vec{E} = 30\sin\left(6\times10^8 t + kz\right)\hat{i}$ then match the following columns.**

	Column I	Column II
A.	Direction of propagation of EM wave	P. −yaxis
B.	Wave length of EM wave	Q. 2 m^{-1}
C.	Direction of magnetic field of EM wave	R. 3.14 m
D.	Propagation constant 'k' of EM wave	S. −zaxis

(1) a-p, b-r, c-s, d-r

(2) a-s, b-r, c-p, d-q

(3) a-p, b-q, c-s, d-s

(4) a-s, b-r, c-p, d-p

NEW LIGHT INSTITUTE All India Full Syllabus Test–31 [Paper]

36. **Resonance occurs in a series LCR circuit when the frequency of the applied emf is 1000 Hz. Then.**

(a) When frequencey = 1200 Hz, then the current through the voltage source will be ahead of emd. of the source

(b) The impedance of the circuit is minimum at f = 1000 Hz.

(c) At resonance the voltage across L and current in C differ in phase by 180^0

(d) If the value of C is doubled resonace occurs at f = 2000 Hz.

(1) only b is correct (2) a, b, c are correct

(3) c & d are correct (4) All are correct

37. **If R, C, and L be the resistance, capacitance and inductance in the circuit in which ac of frequency f is set up, then which of the following have the dimensions of R?**

(a) f C

(b) f L

(c) 1 / f C

(d) L / f

(1) Only a & b (2) Only b & c

(3) Only c & d (4) All

38. **Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).**

Assertion (A) : The diamagnetic substances do not obey Curies's law.

Reason (R) : At curie point ferromagnetic substances start behaving as paramagnetic substances.

In the light of the above statements, choose the most appropriate answer from the options give below:

(1) Both A and R are true and R is the correct explanation of A

(2) Both A and R are true and R is Not the correct explanation of A

(3) A is true but R is false

(4) A is false but R is true.

39. A vernier callipers having 1 main scale division = 0.1 cm is designed to have a least count of 0.02 cm. If n be the number of divisions on vernier scale and m be the length of vernier scale, then:

(1) n = 10, m = 0.5 cm

(2) n = 9, m = 0.4 cm

(3) n = 10, m = 0.8 cm

(4) n = 10, m = 0.2 cm

40. Current sensitiveity of a moving coil galvanometer is 5 div/mA and its voltage sensitivity (angular deflection per unit voltage applied) is 20 div/V. The resistance of the galvanometer is:

(1) 40 Ω (2) 25 Ω

(3) 250 Ω (4) 500 Ω

41. To know the resistance G of a galvanometer by half deflection method, a battery of emf V_E and resistance R is used to deflect the galvanometer by angle θ. If a shunt of resistance S is needed to get half deflection then G, R and S are related by the equation :

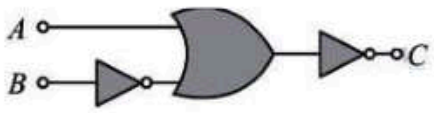
(1) 2S (R+G) = RG

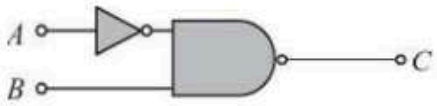
(2) S (R+G) = GR

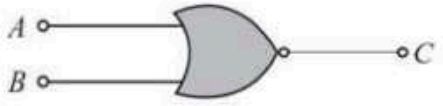
(3) 2G = S

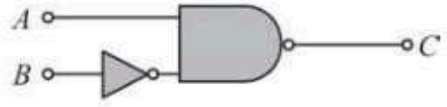
(4) 2S = G

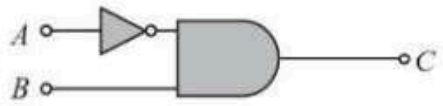
42. The logic circuit shown above is equivalent to :



(1) 

(2) 

(3) 

(4) 

NEW LIGHT INSTITUTE All India Full Syllabus Test–31 [Paper]

43. The speed of a projectile at its maximum height is $\frac{\sqrt{3}}{2}$ times its initial speed. If the range of the projectile is 'P' times the maximum height attained by it, P is:

(1) $\frac{4}{3}$

(2) $2\sqrt{3}$

(3) $4\sqrt{3}$

(4) $\frac{3}{4}$

44. A stone of mass 1 kg tied to a light string of length L = 10 m is whirling in a circular path in vertical plane. If the ratio of maximum to minimum tension in the string is 3, find the speed of the stone at the highest point :

(1) $20\sqrt{2}$ m / s

(2) 20 m/s

(3) 10 m/s

(4) 40 m/s

45. The period of oscillation of a mass m suspended from a spring is 2 seconds. If along with this mass, another mass of 2 kg is also suspended, the period of oscillation increases by one second. The mass m will be:

(1) 2 kg

(2) 1.6 kg

(3) 8 kg

(4) 2.6 kg

TOPIC : FULL SYLLABUS

Atomic Masses : H=1, He=4, C=12, N=14, O
K=39, Ca=40, Fe=56, Cu=63.5, Br=80, Ag

46. **Myoglobin, (Mb), an oxygen storage protein, contains 0.34% Fe by mass and in each molecule of myoglobin one ion of Fe is present. Molar mass of Mb (g mol^{-1}) is :**

(Molar mass of Fe = 55.845 g mol^{-1})

(1) 16407

(2) 164206

(3) 16425

(4) 164250

47. **The reaction of 2,4-hexadiene with one equivalent of bromine at 0°C gives a mixture of two compounds 'X' and 'Y'. If 'X' is 4,5-dibromohex-2-ene, 'Y' is :**

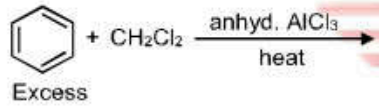
(1) 2,5-dibromohex-2-ene

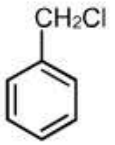
(2) 2,5-dibromohex-3-ene

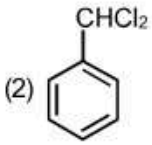
(3) 2,3-dibromohex-3-ene

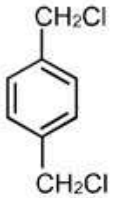
(4) 3,4-dibromohex-3-ene

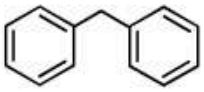
48. **The major product of the following reaction is :**



(1) 

(2) 

(3) 

(4) 

49. **An electrochemical cell was constructed with Fe^{2+}/Fe and Cd^{2+}/Cd at 25°C with initial concentrations of $[Fe^{2+}] = 0.800$ M and $[Cd^{2+}] = 0.250$ M. The EMF of the cell when $[Cd^{2+}]$ becomes 0.100 M is :**

Half cell	E°(V)
$Fe^{2+}(aq) / Fe(s)$	– 0.44
$Cd^{2+}(aq) / Cd(s)$	– 0.40

(1) 0.013 V (2) 0.011 V

(3) 0. 051 V (4) 0.022 V

NEW LIGHT INSTITUTE All India Full Syllabus Test–31 [Paper]

50. **The kinetic energy of the photoelectrons ejected by a metal surface increased from 0.6 eV to 0.9 eV when the energy of the incident photons was increased by 20%. The work function of the metal is :**

(1) 0.66 eV

(2) 0.72 eV

(3) 0.90 eV

(4) 0.30 eV

51. **The correct order of CFSE among $[Zn(NH_3)_4]^{2+}$ and $[Co(NH_3)_6]^{2+}$ and $[Co(NH_3)_6]^{3+}$ is :**

(1) $[Co(NH_3)_6]^{3+} > [Co(NH_3)_6]^{2+} > [Zn(NH_3)_4]^{2+}$

(2) $[Zn(NH_3)_4]^{2+} > [Co(NH_3)_6]^{2+} > [Co(NH_3)_6]^{3+}$

(3) $[Co(NH_3)_6]^{3+} > [Zn(NH_3)_4]^{2+} > [Co(NH_3)_6]^{2+}$

(4) $[Co(NH_3)_6]^{2+} > [Co(NH_3)_6]^{3+} > [Zn(NH_3)_4]^{2+}$

52. **Which of the following is correct about the isoelectronic species, Li^+ and H^- :**

I. **H^- is larger in size than Li^+**

II. **Li^+ is better reducing agent than H^-**

III. **It requires more energy to remove an electron from H^- than from Li^+**

IV. **The chemical properties of the H^- and Li^+ ions are the same.**

(1) I only

(2) II and III only

(3) I, II and IV only

(4) I and II only

53. **Number of products formed (ignoring stereoisomerism) in the monochlorination of ethylcyclohexane is :**

(1) 6

(2) 8

(3) 5

(4) 4

54. **Which of the following pairs of compounds can be stable while retaining the identity of each compound in the pair over a period of time :**

I. **$FeCl_3$, $SnCl_2$** **II.** **$HgCl_2$, $SnCl_2$**

III. **$FeCl_2$, $SnCl_2$** **IV.** **$FeCl_3$, KI**

(1) I only (2) I and III only

(3) III only (4) II and IV only

55. The reaction $xX(g) \rightleftharpoons yY(g) + zZ(g)$ was carried out at a certain temperature with an initial pressure of X = 30 bar. Initially 'Y' and 'Z' were not present. If the equilibrium partial pressure of 'X', 'Y' and 'Z' are 20, 5 and 10 bar respectively x : y : z is :

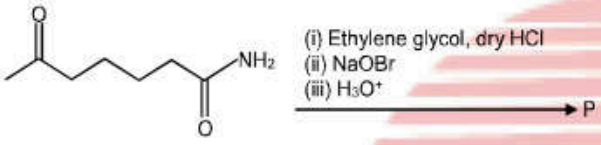
(1) 4 : 1 : 2

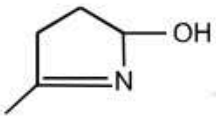
(2) 2 : 1 : 2

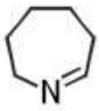
(3) 1 : 2 : 1

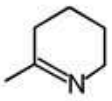
(4) 1 : 1 :2

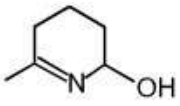
56. The major product 'P' formed in the following sequence of reactions is :



(1) 

(2) 

(3) 

(4) 

57. The unit of Planck's constant, 'h', is the same as that of

(1) angular momentum

(2) energy

(3) wavelength

(4) frequency

58. The set in which all the species are diamagnetic is

(1) B_2, O_2, NO

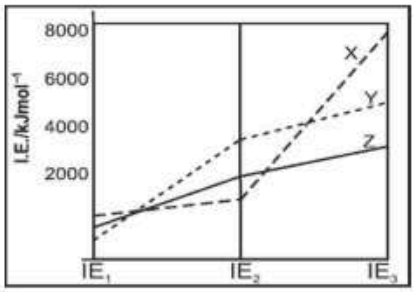
(2) O_2, O_2^+, CO

(3) N_2, O_2^-, CN^-

(4) C_2, O_2^{2-}, NO^+

NEW LIGHT INSTITUTE All India Full Syllabus Test–31 [Paper]

59. The following qualitative plots depict the first, second and third ionization energies (I.E.) of Mg, Al and K. Among the following, the correct match of I.E. and the metal is :



(1) X-Al; Y-Mg ; Z-K

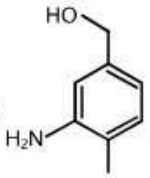
(2) X-Mg; Y-Al ; Z-K

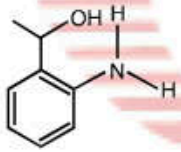
(3) X-Mg; Y-K ; Z-Al

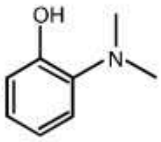
(4) X-Al; Y-K ; Z-Mg

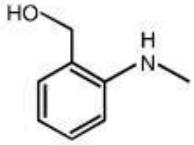
60. The structure of compound 'X' ($C_8H_{11}NO$) based on the following tests and observations is :

Reagent	→	Observation
Neutral $FeCl_3$	→	No coloration
Lucas reagent	→	Turbidity
$NaNO_2$/HCl at 273 K	→	Yellow oil

(1) 

(2) 

(3) 

(4) 

61. The number of stereoisomers is maximum for

(1) $[Co(en)_3]^{3+}$

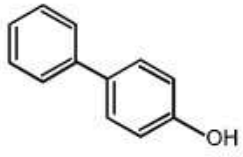
(2) $[Co(en)_2ClBr]^+$

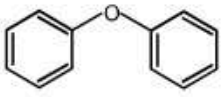
(3) $[Co(NH_3)_4Cl_2]^+$

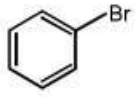
(4) $[Co(NH_3)_4ClBr]^+$

62. Reaction of C_6H_5MgBr with phenol gives :

(1) 

(2) 

(3) 

(4) 

63. The work done (kJ) in the irreversible isothermal compression of 2.0 moles of an ideal gas from 1 bar to 100 bar at 25°C at constant external pressure of 500 bar is :

(1) 2452

(2) 490

(3) 2486

(4) –490

64. An electron present in the third exited state of a H atom returns of the first exited state and then to the ground state. If λ_1 and λ_2 are the wavelengths of light emitted in these two transitions respectively, $\lambda_1 : \lambda_2$ is :

(1) 4 : 1

(2) 5 : 9

(3) 3 : 1

(4) 2 : 1

65. The percentage dissociation of 0.08 M aqueous acetic acid solution at 25°C is (K_a of acetic acid at 25°C = 1.8×10^{-5})

(1) 2.92

(2) 1.5

(3) 1.2

(4) 4.8

66. In which of the following, is a new C-C bond formed in the product :

I. $CH_3CHO \xrightarrow{\text{dil. NaOH}}$

II. $CH_3MgCl + C_2H_5OH \xrightarrow{\text{Heat}}$

III. $CO_2 + CH_3MgBr \xrightarrow{H_3O^+}$

IV. $C_2H_2 + NaNH_2 \xrightarrow{CH_3Br}$

(1) I, III and IV only

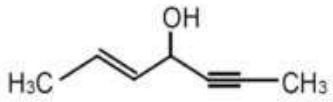
(2) II and III only

(3) III only

(4) III and IV only

NEW LIGHT INSTITUTE All India Full Syllabus Test–31 [Paper]

67. IUPAC name of the following molecule is :



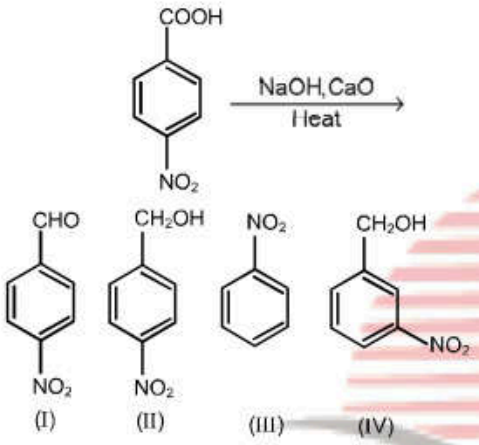
(1) 4-hydroxyhept-2-en-5-yne

(2) hept-2-en-5-yn-4-ol

(3) hept-5-en-2-yn-4-ol

(4) 4-hydroxyhept-5-en-2-yne

68. The product/s of the following reaction is/are



(1) I and II only

(2) II only

(3) III only

(4) IV only

69. The anhydride of HNO_3 is :

(1) NO

(2) NO_2

(3) N_2O

(4) N_2O_5

70. The $F - \hat{X} - F$ bond angle is the smallest in (X is the central atom)

(1) CF_4

(2) NF_3

(3) OF_2

(4) XeF_5^-

71. The correct IUPAC name of the compound, $[Pt(py)_4][Pt(Br)_4]$ is :

(1) tetrapyridineplatinum(II) tetrabromidoplatinate(II)

(2) tetrabromidoplatinum(IV) tetrapyridineplatinate(II)

(3) tetrabromidoplatinate(II) tetrapyridineplatinum(II)

(4) tetrapyridineplatinum(IV) tetrabromidoplatinate(IV)

72. **All four types of carbon (1°, 2°, 3° and 4°) are present in :**



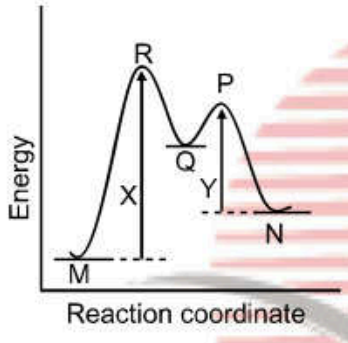
(1) I, II and III only

(2) II, III and IV only

(3) I, II and IV only

(4) II and III only

73. **About the energy level diagram given below, which of the following statement/s is/are correct?**



I. **The reaction is of two steps and 'R' is an intermediate**

II. **The reaction is exothermic and step-2 is rate determining**

III. **'Q' is an intermediate and 'R' is the transition `state for the reaction M→Q**

IV. **'P' is the transition state for the reaction Q→N**

(1) III and IV Only

(2) I, III and IV Only

(3) I, II and IV Only

(4) III only

74. **The mass (g) of NaCl that has to the dissolved to reduce the vapour pressure of 100 g of water by 10% (Molar mass of NaCl = 58.5 g mol^{-1}) is:**

(1) 36.11 g

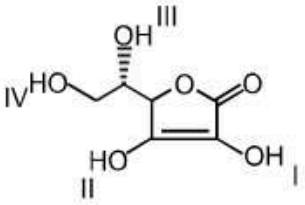
(2) 18.11 g

(3) 81.25 g

(4) 3.61 g

NEW LIGHT INSTITUTE All India Full Syllabus Test–31 [Paper]

75. **The most acidic hydrogen in the following molecule is :**



(1) I

(2) II

(3) III

(4) IV

76. **The correct sequence of reaction which will yield 4-nitrobenzoic acid from benzene is :**

(1) CH_3Cl; HNO_3/H_2SO_4; $KMnO_4/OH^-$

(2) HNO_3/H_2SO_4; $CH_3Cl/AlCl_3$; $KMnO_4/OH^-$

(3) $CH_3Cl/AlCl_3$; $KMnO_4/OH^-$; HNO_3/H_2SO_4

(4) $CH_3Cl/AlCl_3$; HNO_3/H_2SO_4; $KMnO_4/OH^-$

77. **The volume of one drop of aqueous solution from an eyedropper is approximately 0.05 mL. One such drop of 0.2 M HCl is added to 100 mL of distilled water. The pH of the resulting solution will be**

(1) 4.0

(2) 7.0

(3) 3.0

(4) 5.5

78. **In which of the following species the octet rule is NOT obeyed?**

I. I_3^- **II.** N_2O

III. OF_2 **IV.** NO^+

(1) I and IV only

(2) II and III only

(3) I only

(4) IV only

79. **2.0 moles of an ideal gas expands isothermally (27°C) and reversibly from a pressure of 1 bar to 10 bar. The heaviest mass that can be lifted through a height of 10 m by the work of this expansion is**

(1) 50.8 kg

(2) 50.8 g

(3) 117.1 kg

(4) 117.1 g

80. **Which of the following species has one lone pair of electrons on the central atom?**

(1) ClF_3

(2) I_3^-

(3) I_3^+

(4) SF_4

81. **Among the following, the complex ion/s that will have a magnetic moment of 2.82 B.M. is/are**

I. $[Ni(CO)_4]$ **II.** $[NiCl_4]^{2-}$

III. $[Ni(H_2O)_6]^{2+}$ **IV.** $[Ni(CN)_4]^{2-}$

(1) I and IV only

(2) II only

(3) II and III only

(4) II, III and IV only

82. **Morphine, a pain killer is basic with molecular formula $C_{17}H_{19}NO_3$. The conjugate acid of morphine is :**

(1) $C_{17}H_{19}NO_3^+$

(2) $C_{17}H_{18}NO_3$

(3) $C_{17}H_{19}NO_3^-$

(4) $C_{17}H_{20}NO_3^+$

83. **A suboxide of carbon, C_3O_2, has a linear structure. Which of the following is correct about C_3O_2 ?**

I. **Oxidation state of all three C atoms is +2**

II. **Oxidation state of the central C atom is zero**

III. **The molecule contains 4 σ and 4 π bonds**

IV. **Hybridization of the central carbon atom is sp^2**

(1) I and IV only

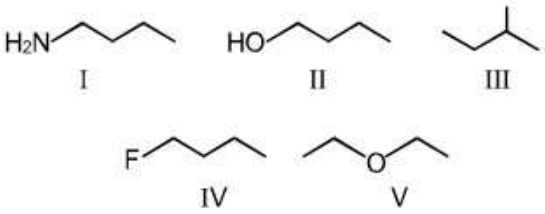
(2) II and III only

(3) II and IV only

(4) III only

NEW LIGHT INSTITUTE All India Full Syllabus Test–31 [Paper]

84. **Among the following, the compounds with highest and lowest boiling points respectively are :**



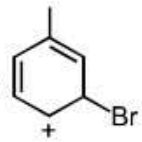
(1) I and III only

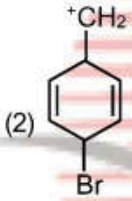
(2) II and III only

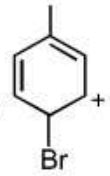
(3) I and IV only

(4) II and V only

85. **The structure that represents the major intermediate formed in bromination of toluene is:**

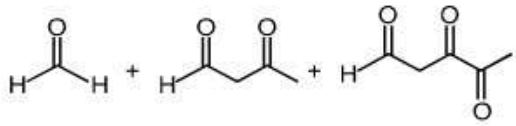
(1) 

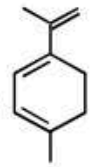
(2) 

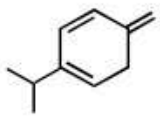
(3) 

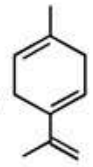
(4) 

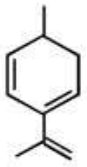
86. **The alkene 'Y' in the following reaction is :**

Alkene Y $\xrightarrow[(ii)\ Me_2S]{(i)\ Ozonolysis}$ 

(1) 

(2) 

(3) 

(4) 

87. **In solid state, PCl_5 exists as $[PCl_4]^+$ $[PCl_6]^-$. The hybridization of P atoms in this solid is/are :**

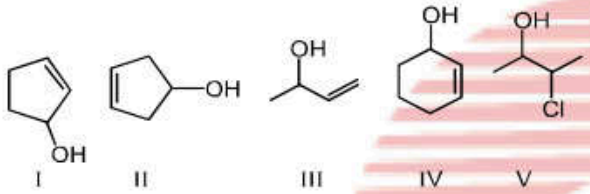
(1) $sp^3d\ (d = d_{x^2-y^2})$

(2) $sp^3d\ (d = d_{z^2})$

(3) sp^3 and $sp^3d^2\ (d = d_{x^2-y^2}, d_{z^2})$

(4) sp^3d and $dsp^3\ (d = d_{z^2})$

88. **Which of the following compounds have chiral carbon atom/s ?**



(1) I and II only

(2) I, III, IV and V

(3) II, IV and V only

(4) II, III and IV only

89. **If the standard electrode potentials of Fe^{3+}/Fe and Fe^{2+}/Fe are –0.04 V and –0.44 V respectively then that of Fe^{3+}/Fe^{2+} is :**

(1) 0.76 V

(2) –0.76 V

(3) 0.40 V

(4) –0.40 V

90. **Given below are two statements.**

Statement I : Arginine and Tryptophan are essential amino acids.

Statement II : Glycine does not have any chiral carbon.

In the light of the above statements, which is the correct option.

(1) Both statement-I and statement-II are correct

(2) Both statement-I and statement-II are incorrect

(3) Statement-I is correct and statement-II is incorrect

(4) Statement-I is incorrect and statement-II is correct

NEW LIGHT INSTITUTE All India Full Syllabus Test–31 [Paper]

TOPIC : FULL SYLLABUS

91. **A prokaryotic organism 'X' that completely lack a cell wall and is smallest living cell known, can survive without oxygen. This organism 'X' will most likely to be**

(1) Nostoc

(2) Methanogens

(3) Mycoplasma

(4) Halophiles

92. **Select the mismatched pair.**

(1) Stilt roots – Maize

(2) Respiratory roots – Rhizophora

(3) Storage roots – Zaminkand

(4) Prop roots – Banyan tree

93. **Following are given two statements, labelled as Assertion (A) and Reason (R).**

Assertion (A): Distribution of bryophytes and pteridophytes is limited.

Reason (R): Pollination occurs by air in bryophytes and pteridophytes.

In the light of the above statements, identify the correct option from the following.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (R) is true but (A) is false

94. **Presence of how many of the features mentioned is/are not considered as fundamental characteristic(s) of chordates?**

Notochord, Dorsal heart, Dorsal solid CNS, Pharyngeal gill-slits, Post-anal tail

Select the correct option.

(1) One

(2) Two

(3) Three

(4) Four

95. Read the statements A and B and choose the most appropriate option.

Statement A: Animals never accumulate ammonia, urea, CO_2, Na^+, K^+, phosphate, sulphate, etc., either by metabolic activities or by excess ingestion.

Statement B: In bony fishes, ammonia is generally excreted by diffusion through gill surfaces as ammonium ions.

(1) Both the statements A and B are true

(2) Only statement A is true

(3) Only statement B is true

(4) Both the statements A and B are false

96. Read the given Assertion (A) and Reason (R) and choose the correct option.

Assertion(A): In roots, pericycle and hypodermis primarily provide mechanical support.

Reason(R): In roots, pericycle is made up of sclerenchyma while hypodermis is made up of collenchymatous cells.

(1) (A) is true but (R) is false

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) Both (A) and (R) are false

(4) Both (A) and (R) are true and (R) is the correct explanation of (A)

97. Listed below are some facts w.r.t. evolution.

a. Hugo de Vries stated that any population has built in variation in characteristics.

b. According to Darwin, the fitness refers ultimately and only to reproductive fitness.

c. Brain of fishes and that of humans are homologous structures.

d. Heart of amphibians and cyclostomes are analogous structures.

Which of the above stated facts are true?

(1) a and b

(2) b and c

(3) a and c

(4) a and d

NEW LIGHT INSTITUTE All India Full Syllabus Test–31 [Paper]

98. Consider the following features

a. Primarily vector borne disease

b. Excess mucus and blood clots in stools

c. Caused by protozoan

d. Spread through faeco-oral route

Which of the above features is/are true for both malaria and amoebic dysentery?

(1) a and d

(2) Only c

(3) Only d

(4) b, c and d

99. Assertion (A) : Pheretima is hermaphrodite and has closed circulatory system

Reason (R): Pheretima are dioecious and perform sexual reproduction.

In the light of above statements, choose the correct option

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

100. Select the common features between the cell organelles also called powerhouse of the cell and photosynthetic plastid.

(a) Double membrane bound

(b) ATP synthesis

(c) Components of endomembrane system

(d) Semi-autonomous organelle

(e) Single stranded DNA

The correct ones are

(1) Only a, b and c

(2) Only b, d and e

(3) Only a, b and d

(4) Only a and d

101.An animal having dorso-ventrally flattened body with absence of alimentary canal is

(1) Filarial worm

(2) Tapeworm

(3) Ascaris

(4) Hookworm

102.Which of the following bones is not a part of rib cage in adult humans?

(1) Thoracic vertebrae

(2) Sternum

(3) False ribs

(4) Cervical vertebrae

103. Find out correct match ?

(1) Metaphase - Decondensation of chromosomes

(2) Anaphase - Splitting of centromere

(3) Telophase - DNA synthesis

(4) S-phase - Chromosomes number become double

104.Which complex of oxidative phosphorylation has two copper centers ?

(1) Complex II

(2) Complex IV

(3) Complex V

(4) Complex III

105.Nuclear envelope consists of parallel membranes with a space called perinuclear space which is about :

(1) 10 to 50 Å

(2) 100 to 500 Å

(3) 1000 to 5000 Å

(4) 1 to 5 Å

106. Match the following columns

Column-I	**Column-II**
(A) P_{680}	(i) reaction centre of PS-I
(B) P_{700}	(ii) reaction centre of PS-II
(C) LHC	(iii) antennae
(D) Plastoquinone	(iv) mobile carrier

(1) A-i, B-ii, C-iv, D-iii

(2) A-ii, B-iv, C-iii, D-i

(3) A-iii, B-ii, C-i, D-iv

(4) A-ii, B-i, C-iii, D-iv

NEW LIGHT INSTITUTE All India Full Syllabus Test–31 [Paper]

107.Assertion (A): The thyroid gland is composed of follicles and stromal tissue.

Reason (R): Fluorine and chlorine is essential for the normal rate of hormone synthesis in the thyroid.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

108.The hypothalamic hormones reach the pituitary gland through :

(1) Hypothalamic duct

(2) Axons

(3) Portal system

(4) Both (2) and (3)

109. Which of the following statements is correct?

(1) Initial clue about chromosomal mechanism of sex determination can be tracked back to the experiments carried out in birds.

(2) Henking discovered X-body but could not explain the significance of X-body.

(3) Large number of insects show ZZ-ZW type of sex determination.

(4) Worker honeybees are produced as a result of parthenogenesis.

110. Germ pores of the pollen grain

(a) Are present on the inner wall of sporoderm

(b) Allow the pollen tube to emerge during germination

(c) Exhibit fascinating array of pattern and design on intine.

(d) Help in pollen preservation

(e) Are prominent apertures on the exine.

Choose the correct option from the following.

(1) (a), (b) and (d) are correct

(2) (b) and (e) are correct

(3) (c), (d) and (e) are incorrect

(4) (a), (b) and (c) are incorrect

111.Which of the following has become ubiquitous in occurrence and causes pollen allergy?

(1) Parthenium

(2) Mirabilis

(3) Oxalis

(4) Vallisneria

112.If starch grain size of pea is considered as the phenotype, the alleles show

(1) Co-dominance

(2) Incomplete dominance

(3) Complete dominance

(4) Pleiotropy

113. Skin colour in humans is

(a) An example of pleiotropy

(b) Seen in two distinct alternate forms

(c) Generally controlled by multiple genes

(d) The darkest when the genotype has all the recessive alleles.

The correct ones is/are

(1) (a), (b) and (d) only

(2) (c) only

(3) (b) and (c) only

(4) (a) and (b) only

114.Choose the incorrect statement among the following.

(1) About 350 mya, fish with stout and strong fins could move on land and go back into water.

(2) Sea weeds and few plants existed probably 320 mya.

(3) The process of how non-cellular aggregates of giant macromolecule evolved into eukaryotic cells is now completely known.

(4) In due course of time, single-celled chemoautotrophic organisms slowly evolved into multicellular life forms.

115.Which of the following statement is not correct w.r.t. racemose inflorescence

(1) Shoot axis continues to grow indefinitely

(2) Younger flowers are present towards the apex and older flower are present at the base

(3) Flowers are borne in basipetal order

(4) It is seen in radish, mustard and lupin

NEW LIGHT INSTITUTE All India Full Syllabus Test–31 [Paper]

116.Which of the following is an incorrect statement regarding HIV infection?

(1) Anti-retroviral drugs can completely cure HIV infection

(2) The virus enters into macrophages where RNA genome of the virus replicates to form viral DNA with the help of enzyme reverse transcriptase.

(3) There is always a time lag between the infection and appearance of AIDS symptoms

(4) This infection does not spread by mere touch

117. Given below are few features:

(i) Body is circular in cross-section

(ii) Three germ layers are present but body cavity is not completely lined by mesoderm

(iii) Lacks digestive system thus, absorbs nutrients from the host directly through body surface

(iv) Have similar level of body organisation as Nereis

How many of the above given features is/are correct for Ancylostoma?

(1) Four

(2) Three

(3) Two

(4) One

118. Comprehend the statements given below:

(a) All transformants are recombinants

(b) All recombinants are transformants

(c) All plasmids carry antibiotic resistance genes

(d) All selectable markers produce colour in the presence of a chromogenic substrate

(e) Restriction enzymes are found in prokaryotes

Choose the option with only correct set.

(1) (a), (b) and (e)

(2) (c) and (e)

(3) (a), (b), (c) and (d)

(4) (b) and (e)

119. Select the function not performed by the hormones of fight or flight.

(1) Glycogenolysis

(2) Lipolysis and proteolysis

(3) Increased respiratory rate

(4) Pupillary constriction

120. Read the statements given below:

Statement A: One of the basic steps in genetically modifying an organism is maintenance of introduced DNA in the host and transfer of the DNA to its progeny.

Statement B: Techniques to alter the chemistry of genetic material to suitably introduce them into respective host organism comes under the umbrella of genetic engineering.

Choose the correct option.

(1) Both statements A and B are incorrect

(2) Both statements A and B are correct

(3) Only statement A is correct

(4) Only statement B is correct

121. Which of the following options is true for coronary artery?

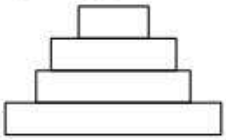
(1) Supplies oxygenated blood to the heart muscle

(2) Supplies deoxygenated blood to lungs

(3) Supplies oxygenated blood to brain

(4) Supplies deoxygenated blood to uterine muscle.

122. Which among the following statements are true for the given ecological pyramid?



a. The food chain representing only one number of tree → herbivorous bird → Hawk can form pyramid of number.

b. It represents pyramid of number of forest ecosystem

c. The length of each bar represents the quantity of parameter taken in consideration

d. In such pyramid, the number of producer is always higher than the number of top carnivore

e. In all types of ecosystems, this can represent the pyramid of energy flow

The correct ones are

(1) a and b only (2) b, c and d only

(3) d and e only (4) c, d and e only

NEW LIGHT INSTITUTE All India Full Syllabus Test–31 [Paper]

123. Match Column I with Column II and choose the correct option

	Column I		Column II
a.	Joseph Priestley	(i)	His experiments involved both aerobic bacteria and eukaryotic organisms like Cladophora
b.	Cornelius van Neil	(ii)	He performed series of experiments that revealed essential role of air in the growth of green plants
c.	T.W. Engelmann	(iii)	He showed that it is only green parts of the plant that could release oxygen
d.	Jan Ingenhousz	(iv)	He inferred that oxygen evolved by the green plant comes from H_2O not from CO_2

(1) a(i), b(ii), c(iv), d(iii) (2) a(iii), b(iv), c(i), d(ii)

(3) a(ii), b(i), c(iv), d(iii) (4) a(ii), b(iv), c(i), d(iii)

124. Read the following statements and select the correct option

Statement A: Microbes are being used to treat sewage by the process of activated sludge formation.

Statement B: Microbes are used to ferment fish, soyabean and bamboo shoots to make foods.

(1) Both statements are correct

(2) Both statements are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

125. How many chromatids will be there in the pachytene stage of a cell, which has 15 tetrads?

(1) 60 (2) 30

(3) 15 (4) 100

126. Sleeping sickness and bovine spongiform encephalopathy are respectively caused by

(1) Trypanosoma and prions

(2) Plasmodium and viroids

(3) Entamoeba and viruses

(4) Paramoecium and virusoids

127.Identify the following statements as true (T) or false (F) and choose the correct option

A. Gas vacuoles are not bound by any unit membrane system

B. The quasi-fluid of lipids in cell membrane enables the lateral movement of proteins within the overall bilayer

C. Position of future cell plate in cells is determined by microtubules.

D. In flagella, the axoneme usually has nine triplets of radially arranged peripheral microtubules and a central hub

	A	B	C	D
(1)	T	F	F	T
(2)	T	T	T	F
(3)	T	F	T	F
(4)	F	T	F	T

128.Select the incorrect option w.r.t. colostrum

(1) Yellowish fluid secreted by the mother during initial days of lactation period

(2) Has abundant amount of immunoglobulins and nutrients

(3) Provides artificially acquired passive immunity to the baby

(4) Protects the infants from acquiring various infections

129.Select the correct match

(1) Restriction enzymes – Nucleases that belong to the category of hydrolases

(2) Agarose compound – Natural monomer extracted from sea weeds

(3) kan^R – Kanamycin resistance encoding gene present in pBR322

(4) rop – Codes for the proteins that affect copy number but not involved in the replication of the plasmid

NEW LIGHT INSTITUTE All India Full Syllabus Test–31 [Paper]

130.Use of PCR can be employed in

a. Archaeogenetics to amplify and extract DNA from fossils to study

b. Detection and diagnosis of various infectious diseases.

c. DNA forensics for crime investigation

Choose the correct option

(1) Only a

(2) Only b and c

(3) Only c

(4) a, b and c

131.During the process of transcription, both the strands of DNA are not copied because

a. They would form two RNA molecules that are complementary to each other

b. It may lead to prevention of translation of RNA into protein

c. Same type of proteins will be produced in higher quantity

d. They lack structural and promoter genes

The correct ones are

(1) b and c only

(2) c and d only

(3) a, c and d only

(4) a and b only

132.Which of the given statements is true about the recessive traits?

(1) They are not expressed in haploid species

(2) They require two copies of the recessive allele to be expressed in diploid species

(3) They are expressed in the heterozygous condition

(4) They can mask the expression of dominant traits

133.Read the statements given below and choose the correct answer

Statement-I : Linked genes always assort independently of one another.

Statement-II : The frequency of recombination between gene pairs on the same chromosome can be used to determine their map distance on the chromosome.

(1) Statement I is true but statement II is false

(2) Statement II is true but statement I is false

(3) Both the statements I and II are true

(4) Both the statements I and II are false

134.Match column I with column II

	Column I	Column II
a.	Sickle - cell anaemia	(i) Occurs due to the presence of defective blood clotting factor in affected individual
b.	α-Thalassemia	(ii) Affected individual lacks liver enzyme phenylalanine hydroxylase
c.	Phenylketonuria	(iii) Defect is caused by transversion mutation of the gene controlling β-chain of haemoglobin.
d.	Haemophilia	(iv) Controlled by two closely linked genes HBA1 and HBA2 on chromosome 16

(1) a(iv), b(ii), c(i), d(iii)

(2) a(iii), b(iv), c(ii), d(i)

(3) a(iii), b(i), c(ii), d(iv)

(4) a(i), b(iii), c(iv), d(ii)

135.Choose the property which is common to both nerve cells and cardiac muscle cells

(1) Absence of nucleus

(2) Excitability

(3) Extensibility

(4) Absence of cell membrane

136.The type of biomacromolecules whose monomeric unit is nucleotide constitute what % of the total cellular mass

(1) 10–15

(2) 5–7

(3) 70–90

(4) 20–30

137.Observe the matches given below and select the correctly matched option.

(1) Marsupial rat and tiger cat — Convergent evolution

(2) Eyes of Octopus and that of dogs — Divergent evolution

(3) Banded anteater and sugar glider — Adaptive radiation

(4) Mouse and Lemur — Convergent evolution

NEW LIGHT INSTITUTE All India Full Syllabus Test–31 [Paper]

138.A person went for an event to Hong Kong via flight, where he came in contact with a person who was diagnosed with COVID few days later. Now, upon his arrival to India, he was found to be asymptomatic but was still put in isolation to prevent the spread of the virus. Which of the following techniques is used to early diagnose whether he is infected with corona virus or not?

(1) Serum analysis

(2) Biopsy

(3) RT-PCR

(4) Urine analysis

139.In the biosphere, heterogeneity can be seen at multiple levels like,

a. Organisms

b. Biomes

c. Species

d. Populations

e. Macromolecules

The correct ones are

(1) Only a and c

(2) Only c and d

(3) All

(4) Only b and d

140. Choose the correct match w.r.t. humans.

(1) Oogenesis — Initiated before the birth of females

(2) Acrosome — Filled with hormones

(3) Antrum — A cavity present within primary follicles

(4) First polar body — Larger than secondary oocyte in size

141.Which of the following characteristics is common in both frogs and cockroaches?

(1) Body is differentiated into head, thorax, neck and abdomen

(2) Gaseous exchange takes place in the region of tracheoles

(3) Presence of red blood cells

(4) Oviparity

142.Upon vasectomy in human males, the recently formed sperms from seminiferous tubules cannot reach

(1) Vasa efferentia

(2) Epididymis

(3) Ejaculatory duct

(4) Rete testes

143. Read the following statements w.r.t. humans:

Statement A: We cannot directly alter the pulmonary volume.

Statement B: The anatomical setup of lungs in thorax is such that any change in the volume of the thoracic cavity will be reflected in the pulmonary cavity.

Statement C: during inspiration volume of thoracic cavity increases and intrapulmonary pressure decreases

Choose the correct option.

(1) All the statements are incorrect

(2) Only statement A is correct

(3) All the statements are correct

(4) Only statement C is incorrect

144. Select incorrect statement.

(1) Families are characterised on the basis of both vegetative and reproductive features of plant species.

(2) Species comprises a group of related genera which has more characters in common in comparison to genera of other species

(3) Convolvulaceae and Solanaceae are included in the order polymoniales

(4) Classes with a few similar characters are assigned to a higher category called division

145. In an ecosystem, 200 kJ of sunlight energy falls on the organism that occupies the first trophic level of a food chain. What amount of this energy will be utilised by the organism for its different biological activities which occupies the third trophic level in this food chain:

(1) 2 J

(2) 1800 J

(3) 200 J

(4) 5000 J

146. a researcher in a lab isolated the protoplasts of both onion and garlic cells and fused them to form hybrids for obtaining characteristics of both, then this process is called :

(1) Genetic cloning

(2) Somatic hybridisation

(3) Micropropagation

(4) Bioprocess engineering

NEW LIGHT INSTITUTE All India Full Syllabus Test–31 [Paper]

147. How many cycles of PCR are required to generate 256 copies of DNA from a 10 kb dsDNA molecule:

(1) 6

(2) 10

(3) 8

(4) 12

148. Read the given statement and choose the correct option to fill in the blank.

lac operon may function constitutively if a mutation occurs in the_____region.

(1) Operator

(2) lac z

(3) Inducer

(4) lac a

149. In which of the following parts of plant. would you expect to find the conjunctive tissue and pericycle:

(1) Leaves of monocots

(2) Leaves of dicots

(3) Stems of grasses

(4) Roots of dicots

150. Read the statements given below and identify correct option.

Statement -I : A fruit which develops from other floral part except ovary is said to be a false fruit.

Statement -II : Some species of Asteraceae family have evolved the mechanism of apomixis.

(1) Both the statement-I and statement-II are true

(2) Both the statement-I and statement-II are false

(3) Statement-I is false butstatement-Ii is true

(4) Statement-I is true but statement-II is false

151. Statement-I:- In frog, eyes and internal ears are well organised structures.

Statement-II- In frog sensory papillae, taste buds and nasal epithelium are cellular aggregation around nerve endings.

(1) Both statement I and statement II are incorrect

(2) Statement I is correct but statement II is incorrect

(3) Statement I is incorrect but statement II is correct

(4) Both statement I and statement II are correct

152. Which structure is tracts of myelinated nerve fibres:

(1) RBCs

(2) Hippocampus

(3) Corpus callosum

(4) Amygdala

153. Which of following statement is not true:

(1) Saheli (oral contraceptive) was developed in Uttar Pradesh state of India.

(2) A rapid increase in IMR and MMR are probable reasons for population explosion.

(3) The world population was around 7.2 billion in 2011.

(4) There is ban on sex determination by amniocentasis.

154. Which of the following composition is called biogas:

(1) CH_4, SO_2, H_2

(2) CH_4, CO_2, H_2

(3) C_2H_6, CO_2, H_2

(4) CH_4, N_2O, H_2

155. Which one of the following statement is correct regarding gel electrophoresis :

(1) Separated bands of DNA are cut from gel, this is known as spooling.

(2) DNA fragments being positively charged move towards anode under electric field.

(3) DNA fragments separate according to their size.

(4) Separated DNA fragments are stained by using methylene blue.

156. Assertion (A) : Cytoplasm is the main arena of cellular activity in plant cells only.

Reason (R) : Various chemical reactions occur in cytoplasm to keep the cell in the living state.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Assertion is False but the Reason is True.

NEW LIGHT INSTITUTE All India Full Syllabus Test–31 [Paper]

157. Which motivates youngsters towards drugs and alcohol use:

(1) Curiosity

(2) Need for adventure and excitement

(3) Experimentation

(4) All of these

158. In the given statements, choose the incorrect one.

(1) Spermatogonia are immature male germ cells

(2) Testis is covered by a dense covering

(3) Each testis has about 250 compartments called rete testis

(4) Germ cells receive nutrition from sertoli cells

159. After spermiogenesis, sperm head becomes embedded in Sertoli cell, and is finally released into the seminiferous tubules by the process called as :

(1) spermiation

(2) semination

(3) spermatogenesis

(4) ejaculation

160. In female Cockroach, which segment is boat shaped :

(1) 6th Tergum

(2) 7th Sternum

(3) 8th Sternum

(4) 9th Sternum

161. Which of the following is not a characteristic of humus:

(1) It is rich in organic matter such as lignin and cellulose.

(2) It is colloidal in nature and serves as a reservoir of nutrients.

(3) It is highly resistant to microbial action and undergoes slow decomposition.

(4) It is further degraded by the process of humification.

162. Which of the following statement is not correct with respect to translation:

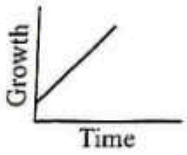
(1) The amino acids are joined by a peptide bond

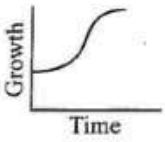
(2) Formation of a peptide bond does not requires energy

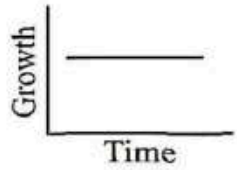
(3) Peptide bond formation is catalysed by 23s-rRNA in bacteria

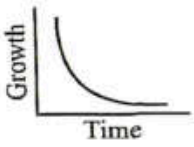
(4) Amino acids are activated in the presence of ATP

163. Which one is the correct graph for arithmetic growth :

(1) 

(2) 

(3) 

(4) 

164. Which of the following is not related to thallus of members of phaeophyceae :

(1) Frond
(2) Stipe
(3) Holdfast
(4) Archegonium

165. Which statement is incorrect regarding predator

(1) Predators act as conduits for energy transfer across trophic level
(2) Predators keep prey population under control
(3) Predators maintain species diversity in a community
(4) Predators always over exploits its prey

166. Biological control method adopted in agricultural pest control are based on :

(1) Commensalism
(2) Amensalism
(3) Parasitism
(4) Predation

167. Which of following statements is incorrect :

(1) Generally, monocotyledonous seeds are endospermic but some as in orchids are non-endospermic.
(2) The egg apparatus consists of one synergid and one egg cell
(3) Fruit is the characteristic of flowering plants
(4) Only a small proportion of plants use abiotic agents in pollination.

NEW LIGHT INSTITUTE All India Full Syllabus Test–31 [Paper]

168. Growth of pollen tube can be measured in terms of:

(1) Increase in volume
(2) Increase in surface area
(3) Increase in length
(4) Increase in cell number

169. What is the position of ovary in the given flower :



(1) Superior
(2) Inferior
(3) Perigynous
(4) Epigynous

170. In Escherichia coli,

(a) Genes are split
(b) Intervening sequences appear in mRNA
(c) Structural gene is polycistronic
(d) Different types of proteins are synthesised by monocistronic gene

Choose the correct statement(s)

(1) (a) and (c) only
(2) (a), (b) and (d) only
(3) All except (d)
(4) (c) only

171. Which of the following have similar mode of action in terms of preventing conception ?

(1) Progestogens injections and steroidal oral contraceptive pills
(2) Implants containing oxytocin and non-steroidal oral contraceptive pills
(3) Barrier contraceptives and non-steroidal oral contraceptive pills
(4) Terminal method of contraception and chemical contraceptives

172. Which of the following statements is false ?

(1) The same yeast used for bread-making, is also used for fermenting malted cereals to produce ethanol

(2) The large holes in 'Swiss cheese' are due to the production of a large amount of CO_2 by a fungus

(3) The dough which is used for making foods such as Dosa and idli is fermented by using bacteria

(4) Lactobacillus, a LAB, play a very beneficial role in checking disease causing microbes in our stomach

173.Identify the incorrectly matched pair

(1) Over exploitation – Stellar's sea cow and passenger pigeon

(2) Evil quartet – Habitat loss and fragmentation

(3) Co-extinction – Fig and fig wasp

(4) Nile perch introduced in Lake Victoria – Extinction of Clarias gariepinus

174. The whole plant can be regenerated from any part of a plant, when grown in a test tube under sterile conditions in special nutrient media. This nutrient medium must contain all of the following except

(1) A carbon source

(2) Growth regulators

(3) Vitamins

(4) Enzymes

175. Select the correct option for given diagram



(A) Pollination by water

(B) Pollination by insect

(C) Two kinds of spore

(D) Vascular plant

(1) B, C, D

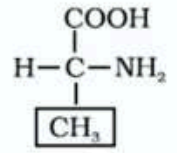
(2) C, D

(3) A, C, D

(4) D only

NEW LIGHT INSTITUTE All India Full Syllabus Test–31 [Paper]

176. Select the correct option for given diagram



(A) Monomer of homopolymer

(B) Monomer of heteropolymer

(C) Formation of zwitter ion

(D) acidic compound

(E) basic compound

(1) B, C, D

(2) B, C

(3) A, C, D

(4) A, D, E

177.Select the correct option for given reaction

$$C_6H_{12}O_6 + O_2 \rightarrow 2C_3H_4O_3 + 2H_2O$$

A. Enzyme mediated

B. Glycolysis

C. Catabolic

D. Respiratory process

E. Occurs in Streptococcus

F. Occurs in Spirogyra

(1) A, B, C, D, E only

(2) B, C, D, E only

(3) B, C, E only

(4) All

178. Select the on-site conservation

(1) Tissue culture

(2) Botanical garden

(3) Sacred grove

(4) Seed bank

179.In a given sample of RNA, there are 666 bases, the code for a protein with 222 amino acids and the base at position 601 is deleted such that length of RNA now becomes 665 base long. How many codons will be altered ?

(1) 220

(2) 22

(3) 222

(4) 11

180. If number of days of menstrual cycle are 35 days in a female then what will be the ovulation day in that female ?

(1) 20th days

(2) 21st days

(3) 49th days

(4) 14th days

NEW LIGHT INSTITUTE All India Full Syllabus Test–31 [Paper]

Space for

Space for I

